



as3880=> \i ss/q00.sql

```
SELECT sequence_number, as_of_year, respondent_id, agency_name, agency_abbr FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----
sequence_number | 1
as_of_year      | 2017
respondent_id   | 62-1532940
agency_name     | Department of Housing and Urban Development
agency_abbr     | HUD

-[ RECORD 2 ]-----
sequence_number | 2
as_of_year      | 2017
respondent_id   | 80-0743699
agency_name     | Department of Housing and Urban Development
agency_abbr     | HUD

-[ RECORD 3 ]-----
sequence_number | 3
as_of_year      | 2017
respondent_id   | 0000177957
agency_name     | Federal Reserve System
agency_abbr     | FRS
```

as3880=>



s/q01.sqas3880=> \i ss/q01.sql

```
SELECT sequence_number, agency_code, loan_type_name, loan_type, property_type_name
FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----+-----
sequence_number      | 1
agency_code          | 7
loan_type_name        | Conventional
loan_type             | 1
property_type_name    | One-to-four family dwelling (other than manufactured housing)
-[ RECORD 2 ]-----+-----
sequence_number      | 2
agency_code          | 7
loan_type_name        | FHA-insured
loan_type             | 2
property_type_name    | One-to-four family dwelling (other than manufactured housing)
-[ RECORD 3 ]-----+-----
sequence_number      | 3
agency_code          | 2
loan_type_name        | Conventional
loan_type             | 1
property_type_name    | One-to-four family dwelling (other than manufactured housing)
```

as3880=>



as3880=> \i ss/q02.sql

```
SELECT sequence_number, property_type, loan_purpose_name, loan_purpose, owner_occu  
pancy_name FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----+-----  
sequence_number      | 1  
property_type        | 1  
loan_purpose_name      | Home purchase  
loan_purpose           | 1  
owner_occupancy_name | Owner-occupied as a principal dwelling  
-[ RECORD 2 ]-----+-----  
sequence_number      | 2  
property_type        | 1  
loan_purpose_name      | Home purchase  
loan_purpose           | 1  
owner_occupancy_name | Owner-occupied as a principal dwelling  
-[ RECORD 3 ]-----+-----  
sequence_number      | 3  
property_type        | 1  
loan_purpose_name      | Refinancing  
loan_purpose           | 3  
owner_occupancy_name | Owner-occupied as a principal dwelling
```

as3880=>



as3880=> \i ss/q03.sql

```
SELECT sequence_number, owner_occupancy, loan_amount_000s, preapproval_name, preap  
proval FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
owner_occupancy	1
loan_amount_000s	272
preapproval_name	Not applicable
preapproval	3

-[RECORD 2]-----+-----

sequence_number	2
owner_occupancy	1
loan_amount_000s	151
preapproval_name	Preapproval was not requested
preapproval	2

-[RECORD 3]-----+-----

sequence_number	3
owner_occupancy	1
loan_amount_000s	192
preapproval_name	Not applicable
preapproval	3

as3880=>



as3880=> \i ss/q04.sql

```
SELECT sequence_number, action_taken_name, action_taken, msamd_name, msamd FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
action_taken_name	Loan purchased by the institution
action_taken	6
msamd_name	Camden - NJ
msamd	15804

-[RECORD 2]-----+-----

sequence_number	2
action_taken_name	Application withdrawn by applicant
action_taken	4
msamd_name	
msamd	

-[RECORD 3]-----+-----

sequence_number	3
action_taken_name	Application withdrawn by applicant
action_taken	4
msamd_name	Ocean City - NJ
msamd	36140

as3880=>



as3880=> \i ss/q05.sql

```
SELECT sequence_number, state_name, state_abbr, state_code, county_name FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----

sequence_number	1
state_name	New Jersey
state_abbr	NJ
state_code	34
county_name	Camden County

-[RECORD 2]-----

sequence_number	2
state_name	New Jersey
state_abbr	NJ
state_code	34
county_name	

-[RECORD 3]-----

sequence_number	3
state_name	New Jersey
state_abbr	NJ
state_code	34
county_name	Cape May County

as3880=>



```
/q06.sqas3880=> \i ss/q06.sql
```

```
SELECT sequence_number, county_code, census_tract_number, applicant_ethnicity_name  
, applicant_ethnicity FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----+-----
```

sequence_number	1
county_code	7
census_tract_number	6035.06
applicant_ethnicity_name	Not applicable
applicant_ethnicity	4

```
-[ RECORD 2 ]-----+-----
```

sequence_number	2
county_code	
census_tract_number	
applicant_ethnicity_name	Information not provided by applicant in mail, Internet , or telephone application
applicant_ethnicity	3

```
-[ RECORD 3 ]-----+-----
```

sequence_number	3
county_code	9
census_tract_number	0218.04
applicant_ethnicity_name	Not Hispanic or Latino
applicant_ethnicity	2

```
as3880=>
```



as3880=> \i ss/q07.sql

```
SELECT sequence_number, co_applicant_ethnicity_name, co_applicant_ethnicity, appli
cant_race_name_1, applicant_race_1 FROM Preliminary ORDER BY sequence_number LIMIT
 3;
```

-[RECORD 1]-----+

sequence_number	1
co_applicant_ethnicity_name	Not applicable
co_applicant_ethnicity	4
applicant_race_name_1	Not applicable
applicant_race_1	7

-[RECORD 2]-----+

sequence_number	2
co_applicant_ethnicity_name	Information not provided by applicant in mail, Inter
net, or telephone application	
co_applicant_ethnicity	3
applicant_race_name_1	Information not provided by applicant in mail, Inter
net, or telephone application	
applicant_race_1	6

-[RECORD 3]-----+

sequence_number	3
co_applicant_ethnicity_name	Not Hispanic or Latino
co_applicant_ethnicity	2
applicant_race_name_1	White
applicant_race_1	5

as3880=>



as3880=> \i ss/q08.sql

```
SELECT sequence_number, applicant_race_name_2, applicant_race_2, applicant_race_name_3, applicant_race_3 FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+--

sequence_number	1
applicant_race_name_2	
applicant_race_2	
applicant_race_name_3	
applicant_race_3	

-[RECORD 2]-----+--

sequence_number	2
applicant_race_name_2	
applicant_race_2	
applicant_race_name_3	
applicant_race_3	

-[RECORD 3]-----+--

sequence_number	3
applicant_race_name_2	
applicant_race_2	
applicant_race_name_3	
applicant_race_3	

as3880=>



as3880=> \i ss/q09.sql

```
SELECT sequence_number, applicant_race_name_4, applicant_race_4, applicant_race_name_5, applicant_race_5 FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+--

sequence_number	1
applicant_race_name_4	
applicant_race_4	
applicant_race_name_5	
applicant_race_5	

-[RECORD 2]-----+--

sequence_number	2
applicant_race_name_4	
applicant_race_4	
applicant_race_name_5	
applicant_race_5	

-[RECORD 3]-----+--

sequence_number	3
applicant_race_name_4	
applicant_race_4	
applicant_race_name_5	
applicant_race_5	

as3880=>



as3880=> \i ss/q10.sql

```
SELECT sequence_number, co_applicant_race_name_1, co_applicant_race_1, co_applicant_race_name_2, co_applicant_race_2 FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
co_applicant_race_name_1	Not applicable
co_applicant_race_1	7
co_applicant_race_name_2	
co_applicant_race_2	

-[RECORD 2]-----+-----

sequence_number	2
co_applicant_race_name_1	Information not provided by applicant in mail, Internet, or telephone application
co_applicant_race_1	6
co_applicant_race_name_2	
co_applicant_race_2	

-[RECORD 3]-----+-----

sequence_number	3
co_applicant_race_name_1	White
co_applicant_race_1	5
co_applicant_race_name_2	
co_applicant_race_2	

as3880=>



as3880=> \i ss/q11.sql

```
SELECT sequence_number, co_applicant_race_name_3, co_applicant_race_3, co_applicant_race_name_4, co_applicant_race_4 FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+--

sequence_number	1
-----------------	---

co_applicant_race_name_3	
--------------------------	--

co_applicant_race_3	
---------------------	--

co_applicant_race_name_4	
--------------------------	--

co_applicant_race_4	
---------------------	--

-[RECORD 2]-----+--

sequence_number	2
-----------------	---

co_applicant_race_name_3	
--------------------------	--

co_applicant_race_3	
---------------------	--

co_applicant_race_name_4	
--------------------------	--

co_applicant_race_4	
---------------------	--

-[RECORD 3]-----+--

sequence_number	3
-----------------	---

co_applicant_race_name_3	
--------------------------	--

co_applicant_race_3	
---------------------	--

co_applicant_race_name_4	
--------------------------	--

co_applicant_race_4	
---------------------	--

as3880=>



as3880=> \i ss/q12.sql

```
SELECT sequence_number, co_applicant_race_name_5, co_applicant_race_5, applicant_s  
ex_name, applicant_sex FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
co_applicant_race_name_5	
co_applicant_race_5	
applicant_sex_name	Not applicable
applicant_sex	4

-[RECORD 2]-----+-----

sequence_number	2
co_applicant_race_name_5	
co_applicant_race_5	
applicant_sex_name	Information not provided by applicant in mail, Internet , or telephone application
applicant_sex	3

-[RECORD 3]-----+-----

sequence_number	3
co_applicant_race_name_5	
co_applicant_race_5	
applicant_sex_name	Female
applicant_sex	2

as3880=>



as3880=> \i ss/q13.sql

```
SELECT sequence_number, co_applicant_sex_name, co_applicant_sex, applicant_income_000s, purchaser_type_name FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
co_applicant_sex_name	Not applicable
co_applicant_sex	4
applicant_income_000s	
purchaser_type_name	Fannie Mae (FNMA)

-[RECORD 2]-----+-----

sequence_number	2
co_applicant_sex_name	Information not provided by applicant in mail, Internet, or telephone application
co_applicant_sex	3
applicant_income_000s	70
purchaser_type_name	Loan was not originated or was not sold in calendar year covered by register

-[RECORD 3]-----+-----

sequence_number	3
co_applicant_sex_name	Female
co_applicant_sex	2
applicant_income_000s	47
purchaser_type_name	Loan was not originated or was not sold in calendar year covered by register

as3880=>



as3880=> \i ss/q14.sql

```
SELECT sequence_number, purchaser_type, denial_reason_name_1, denial_reason_1, denial_reason_name_2 FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+--

sequence_number	1
purchaser_type	1
denial_reason_name_1	
denial_reason_1	
denial_reason_name_2	

-[RECORD 2]-----+--

sequence_number	2
purchaser_type	0
denial_reason_name_1	
denial_reason_1	
denial_reason_name_2	

-[RECORD 3]-----+--

sequence_number	3
purchaser_type	0
denial_reason_name_1	
denial_reason_1	
denial_reason_name_2	

as3880=>



sqas3880=> \i ss/q15.sql

```
SELECT sequence_number, denial_reason_2, denial_reason_name_3, denial_reason_3, rate_spread FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+--

sequence_number	1
denial_reason_2	
denial_reason_name_3	
denial_reason_3	
rate_spread	

-[RECORD 2]-----+--

sequence_number	2
denial_reason_2	
denial_reason_name_3	
denial_reason_3	
rate_spread	

-[RECORD 3]-----+--

sequence_number	3
denial_reason_2	
denial_reason_name_3	
denial_reason_3	
rate_spread	

as3880=>



as3880=> \i ss/q16.sql

```
SELECT sequence_number, hoepa_status_name, hoepa_status, lien_status_name, lien_status FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
hoepa_status_name	Not a HOEPA loan
hoepa_status	2
lien_status_name	Not applicable
lien_status	4

-[RECORD 2]-----+-----

sequence_number	2
hoepa_status_name	Not a HOEPA loan
hoepa_status	2
lien_status_name	Secured by a first lien
lien_status	1

-[RECORD 3]-----+-----

sequence_number	3
hoepa_status_name	Not a HOEPA loan
hoepa_status	2
lien_status_name	Secured by a first lien
lien_status	1

as3880=>



as3880=> \i ss/q17.sql

```
SELECT sequence_number, edit_status_name, edit_status, population, minority_popula
tion FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

-[RECORD 1]-----

sequence_number	1
edit_status_name	
edit_status	
population	6278
minority_population	27.829999923706055

-[RECORD 2]-----

sequence_number	2
edit_status_name	
edit_status	
population	
minority_population	

-[RECORD 3]-----

sequence_number	3
edit_status_name	
edit_status	
population	5734
minority_population	15.15999984741211

as3880=>



as3880=> \i ss/q18.sql

```
SELECT sequence_number, hud_median_family_income, tract_to_msamd_income, number_of_
_owner_occupied_units, number_of_1_to_4_family_units FROM Preliminary ORDER BY seq
uence_number LIMIT 3;
```

-[RECORD 1]-----+-----

sequence_number	1
hud_median_family_income	91500
tract_to_msamd_income	144.7100067138672
number_of_owner_occupied_units	1975
number_of_1_to_4_family_units	2018

-[RECORD 2]-----+-----

sequence_number	2
hud_median_family_income	
tract_to_msamd_income	
number_of_owner_occupied_units	
number_of_1_to_4_family_units	

-[RECORD 3]-----+-----

sequence_number	3
hud_median_family_income	73400
tract_to_msamd_income	68.94000244140625
number_of_owner_occupied_units	1728
number_of_1_to_4_family_units	3303

as3880=>



as3880@basic: ~/cs336g28/sql



```
as3880=> \i ss/q19.sql
```

```
SELECT sequence_number, application_date_indicator FROM Preliminary ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----+--
```

```
sequence_number      | 1
```

```
application_date_indicator |
```

```
-[ RECORD 2 ]-----+--
```

```
sequence_number      | 2
```

```
application_date_indicator |
```

```
-[ RECORD 3 ]-----+--
```

```
sequence_number      | 3
```

```
application_date_indicator |
```

```
as3880=>
```



```
<_race_5 IS NOT NULL ORDER BY sequence_number LIMIT 2;  
SELECT sequence_number, applicant_race_name_2, applicant_race_2, applicant_race_name_3,  
applicant_race_3, applicant_race_name_4, applicant_race_4, applicant_race_name_5,  
applicant_race_5 FROM Preliminary WHERE applicant_race_5 IS NOT NULL ORDER BY  
sequence_number LIMIT 2;
```

```
-[ RECORD 1 ]-----+-----
```

sequence_number		2147
applicant_race_name_2		Asian
applicant_race_2		2
applicant_race_name_3		Black or African American
applicant_race_3		3
applicant_race_name_4		Native Hawaiian or Other Pacific Islander
applicant_race_4		4
applicant_race_name_5		White
applicant_race_5		5

```
-[ RECORD 2 ]-----+-----
```

sequence_number		3007
applicant_race_name_2		Asian
applicant_race_2		2
applicant_race_name_3		Black or African American
applicant_race_3		3
applicant_race_name_4		Native Hawaiian or Other Pacific Islander
applicant_race_4		4
applicant_race_name_5		White
applicant_race_5		5

```
as3880=>
```



```
<co_applicant_race_5 IS NOT NULL ORDER BY sequence_number LIMIT 2;  
SELECT sequence_number, co_applicant_race_name_2, co_applicant_race_2, co_applicant_race_name_3, co_applicant_race_3, co_applicant_race_name_4, co_applicant_race_4, co_applicant_race_name_5, co_applicant_race_5 FROM Preliminary WHERE co_applicant_race_5 IS NOT NULL ORDER BY sequence_number LIMIT 2;
```

```
-[ RECORD 1 ]-----+-----
```

sequence_number	2147
co_applicant_race_name_2	Asian
co_applicant_race_2	2
co_applicant_race_name_3	Black or African American
co_applicant_race_3	3
co_applicant_race_name_4	Native Hawaiian or Other Pacific Islander
co_applicant_race_4	4
co_applicant_race_name_5	White
co_applicant_race_5	5

```
-[ RECORD 2 ]-----+-----
```

sequence_number	189318
co_applicant_race_name_2	Asian
co_applicant_race_2	2
co_applicant_race_name_3	Black or African American
co_applicant_race_3	3
co_applicant_race_name_4	Native Hawaiian or Other Pacific Islander
co_applicant_race_4	4
co_applicant_race_name_5	White
co_applicant_race_5	5

```
as3880=>
```



```
<nial_reason_3 IS NOT NULL ORDER BY sequence_number LIMIT 3;  
SELECT sequence_number, denial_reason_name_1, denial_reason_1, denial_reason_name_  
2, denial_reason_2, denial_reason_name_3, denial_reason_3 FROM Preliminary WHERE d  
enial_reason_3 IS NOT NULL ORDER BY sequence_number LIMIT 3;
```

```
-[ RECORD 1 ]-----+-----
```

sequence_number	909
denial_reason_name_1	Collateral
denial_reason_1	4
denial_reason_name_2	Debt-to-income ratio
denial_reason_2	1
denial_reason_name_3	Credit history
denial_reason_3	3

```
-[ RECORD 2 ]-----+-----
```

sequence_number	968
denial_reason_name_1	Credit history
denial_reason_1	3
denial_reason_name_2	Unverifiable information
denial_reason_2	6
denial_reason_name_3	Other
denial_reason_3	9

```
-[ RECORD 3 ]-----+-----
```

sequence_number	998
denial_reason_name_1	Credit history
denial_reason_1	3
denial_reason_name_2	Collateral
denial_reason_2	4
denial_reason_name_3	Other
denial_reason_3	9

```
as3880=>
```



```
<E rate_spread <> '' ORDER BY sequence_number LIMIT 4;  
SELECT sequence_number, rate_spread, edit_status, application_date_indicator FROM  
Preliminary WHERE rate_spread <> '' ORDER BY sequence_number LIMIT 4;
```

```
-[ RECORD 1 ]-----+-----
```

sequence_number	18
rate_spread	01.90
edit_status	
application_date_indicator	

```
-[ RECORD 2 ]-----+-----
```

sequence_number	33
rate_spread	04.14
edit_status	
application_date_indicator	

```
-[ RECORD 3 ]-----+-----
```

sequence_number	69
rate_spread	04.47
edit_status	
application_date_indicator	

```
-[ RECORD 4 ]-----+-----
```

sequence_number	73
rate_spread	01.50
edit_status	
application_date_indicator	

```
as3880=>
```



```
< row_count, min(sequence_number), max(sequence_number) FROM Preliminary;
```

```
! rm -f reconstructed.csv
```

```
\i 02_export_to_csv.sql
```

```
row_count | min | max
```

```
-----+-----+-----
```

```
349563 | 1 | 349563
```

```
(1 row)
```

```
as3880=> \! rm -f reconstructed.csv
```

```
as3880=> \i 02_export_to_csv.sql
```

```
\! ls -l hmda_2017_nj_all-records_labels.csv reconstructed.csv
```

```
\! diff hmda_2017_nj_all-records_labels.csv reconstructed.csv && echo 'diff: no differences - files are identical'
```

```
COPY 349563
```

```
as3880=> \! ls -l hmda_2017_nj_all-records_labels.csv reconstructed.csv
```

```
-rw-r--r-- 1 as3880 as3880 278834016 Mar 18 2019 hmda_2017_nj_all-records_labels.csv
```

```
-rw-r----- 1 as3880 as3880 278834016 Sep 30 22:40 reconstructed.csv
```

```
< && echo 'diff: no differences - files are identical'
```

```
diff: no differences - files are identical
```

```
as3880=> █
```